Chapter 3.1 Inferences for a Mean Vector

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Plan

Inference Concerning a Mean I

$$X_1, ..., X_n \overset{i.i.d}{\sim} \mathcal{N}\left(\mu, \sigma^2\right)$$

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2 \qquad \overline{X}, S^2 \text{independent}$$

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \Rightarrow \quad Z = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$$

$$\frac{(n-1)\,S^2}{\sigma^2} \sim \chi^2_{n-1} \qquad \frac{Z}{\sqrt{\chi^2_{df}/df}} \sim t_{(df)} \qquad \text{if } Z \perp \chi^2_{df}$$

$$\Rightarrow \quad T = \frac{\left[\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}\right]}{\sqrt{\frac{(n-1)S^2}{\sigma^2}\Big/(n-1)}} = \frac{\overline{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$$

$$\Rightarrow \quad P\left(-t_{n-1}\left(\frac{\alpha}{2}\right) \le \frac{\overline{X}-\mu}{S/\sqrt{n}} \le t_{n-1}\left(\frac{\alpha}{2}\right)\right) = 1-\alpha.$$

Inference Concerning a Mean II

Sample Measurements:

$$x_1, \ldots, x_n \qquad \overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2.$$

$$\text{Hypotheses:} \begin{cases} H_0 : \mu = \mu_0 \\ H_A : \mu \neq \mu_0 \end{cases}$$

$$\text{Test Statistic:} t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$\text{RR} : |t_{obs}| \geq t_{n-1}\left(\frac{\alpha}{2}\right) \qquad p_{value} = 2P\left(t_{n-1} \geq |t_{obs}|\right)$$

Squaring t_{obs}:

Reject H_0 if $t_{obs}^2 = n(\overline{x} - \mu_0)\left(s^2\right)^{-1}(\overline{x} - \mu_0) \geq t_{n-1}^2\left(\frac{\alpha}{2}\right) = F_{1,n-1}(\alpha)$.

$(1-\alpha)\,100\%$ Confidence Interval for μ : $\quad \overline{x} \pm t_{n-1}\left(\frac{\alpha}{2}\right)\frac{s}{\sqrt{n}}$

Inference for a Mean Vector – Hotelling's T^2 - I

$$\mathbf{X}_1, ..., \mathbf{X}_n \overset{i.i.d}{\sim} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$\mathbf{X}_j = \begin{bmatrix} X_{j1} \\ \vdots \\ X_{jp} \end{bmatrix} \qquad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_p \end{bmatrix} \qquad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{p1} & \cdots & \sigma_{pp} \end{bmatrix}$$

$$\overline{\mathbf{X}} = \frac{1}{n}\sum_{j=1}^{n} \mathbf{X}_j \qquad \mathbf{S} = \frac{1}{n-1}\sum_{j=1}^{n} \left(\mathbf{X}_j - \overline{\mathbf{X}}\right)\left(\mathbf{X}_j - \overline{\mathbf{X}}\right)'$$

$$(n-1)\,\mathbf{S} \sim \text{Wishart}_{p,n-1}(\boldsymbol{\Sigma}) \equiv \mathbf{W}_{p,n-1} \qquad \overline{\mathbf{X}},\ \mathbf{S} \text{ independent}$$

Under H_0: $\boldsymbol{\mu} = \boldsymbol{\mu}_0$: $\quad \sqrt{n}\left(\overline{\mathbf{X}} - \boldsymbol{\mu}_0\right) \sim \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma})$

$$T^2 = \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma})'\left[\frac{\mathbf{W}_{p,df}}{df}\right]^{-1}\mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma}) \sim \frac{df \times p}{n-p} F_{p,n-p} \text{ when } \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma}) \perp \mathbf{W}_{p,df}.$$

$$df = n-1 \quad \Rightarrow n\left(\overline{\mathbf{X}} - \boldsymbol{\mu}\right)'\mathbf{S}^{-1}\left(\overline{\mathbf{X}} - \boldsymbol{\mu}\right) \sim \left(\tfrac{(n-1)p}{n-p}\right) F_{p,n-p}$$

$$\Rightarrow \quad \left(\tfrac{n-p}{(n-1)p}\right) T^2 \sim F_{p,n-p}.$$

Inference for a Mean Vector – Hotelling's T^2 - II

Sample Measurements:

$$\mathbf{x}_1, \ldots, \mathbf{x}_n \qquad \bar{\mathbf{x}} = \frac{1}{n}\sum_{j=1}^{n} \mathbf{x}_j \qquad \mathbf{S} = \frac{1}{n-1}\sum_{j=1}^{n} (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})'$$

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0 \quad \text{vs} \quad H_A : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$$

Test Statistic: $T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)' \mathbf{S}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$

$$RR : T^2 \geq \left(\frac{(n-1)p}{n-p}\right) F_{p,n-p}(\alpha) \text{ equivalently: } \left(\frac{n-p}{(n-1)p}\right) T^2 \geq F_{p,n-p}(\alpha)$$

$$p_{value} = P\left(F_{p,n-p} \geq \left(\frac{n-p}{(n-1)p}\right) T^2\right)$$

Inference for a Mean Vector – Hotelling's T^2 - III

T^2 is invariant to (full rank) linear transformations of $\underset{p\times 1}{\mathbf{X}}$:

$$\underset{p\times 1}{\mathbf{Y}} = \underset{p\times p}{\mathbf{C}}\,\underset{p\times 1}{\mathbf{X}} + \underset{p\times 1}{\mathbf{d}} \qquad \mathrm{rank}\,(\mathbf{C}) = p$$

$$E\{\mathbf{Y}\} = \boldsymbol{\mu}_{\mathbf{Y}} = \mathbf{C}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{d} \qquad V\{\mathbf{Y}\} = \boldsymbol{\Sigma}_{\mathbf{Y}} = \mathbf{C}\boldsymbol{\Sigma}_{\mathbf{Y}}\mathbf{C}'$$

$$\boldsymbol{\mu}_{\mathbf{Y}0} = \mathbf{C}\boldsymbol{\mu}_0 + \mathbf{d} \qquad \overline{\mathbf{y}} = \mathbf{C}\overline{\mathbf{x}} + \mathbf{d}$$

$$\mathbf{S}_{\mathbf{y}} = \frac{1}{n-1}\sum_{j=1}^{n} \mathbf{C}\left(\mathbf{x}_j - \overline{\mathbf{x}}\right)\left(\mathbf{C}\left(\mathbf{x}_j - \overline{\mathbf{x}}\right)\right)' = \mathbf{CSC}'$$

$$\Rightarrow \quad T_y^2 = n\left(\left(\mathbf{C}\overline{\mathbf{x}} + \mathbf{d}\right) - \left(\mathbf{C}\boldsymbol{\mu}_0 + \mathbf{d}\right)\right)'(\mathbf{CSC}')^{-1}\left(\left(\mathbf{C}\overline{\mathbf{x}} + \mathbf{d}\right) - \left(\mathbf{C}\boldsymbol{\mu}_0 + \mathbf{d}\right)\right) =$$

$$n\left(\overline{\mathbf{x}} - \boldsymbol{\mu}_0\right)'\mathbf{C}'(\mathbf{C}')^{-1}\mathbf{S}^{-1}\mathbf{C}^{-1}\mathbf{C}\left(\overline{\mathbf{x}} - \boldsymbol{\mu}_0\right) = n\left(\overline{\mathbf{x}} - \boldsymbol{\mu}_0\right)'\mathbf{S}^{-1}\left(\overline{\mathbf{x}} - \boldsymbol{\mu}_0\right) = T^2$$

T^2 and Likelihood Ratio Tests

$\mathbf{X}_1, ..., \mathbf{X}_n \overset{i.i.d}{\sim} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ Hypotheses: $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ vs $H_A : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$

Sample Measurements: : $\mathbf{x}_1, ..., \mathbf{x}_n$

Maximum Likelihood Estimate of $\boldsymbol{\Sigma}$: $\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum_{j=1}^{n}(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})'$

Under null hypothesis: $\hat{\boldsymbol{\Sigma}}_0 = \frac{1}{n}\sum_{j=1}^{n}(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)'$

Likelihood Ratio: $\Lambda = \frac{\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})}{\max_{\boldsymbol{\mu},\boldsymbol{\Sigma}} L(\boldsymbol{\mu}, \boldsymbol{\Sigma})} = \left(\frac{|\hat{\boldsymbol{\Sigma}}_0|}{|\hat{\boldsymbol{\Sigma}}|}\right)^{-n/2} = \left(\frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|}\right)^{n/2}$

Wilks' Lambda:

$$\Lambda^{2/n} = \frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|} = \left(1 + \frac{T^2}{n-1}\right)^{-1} \Rightarrow \quad T^2 = \frac{(n-1)\left|\sum_{j=1}^{n}(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)'\right|}{\left|\sum_{j=1}^{n}(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})'\right|} - (n-1)$$

Confidence Regions for Mean Vector - I

$$n\left(\overline{\mathbf{X}}-\boldsymbol{\mu}\right)'\mathbf{S}^{-1}\left(\overline{\mathbf{X}}-\boldsymbol{\mu}\right) \sim \left[\frac{p\,(n-1)}{n-p}\right] F_{p,n-p}$$

$$\Rightarrow \quad P\left(\left[\frac{n-p}{p\,(n-1)}\right] n\left(\overline{\mathbf{X}}-\boldsymbol{\mu}\right)'\mathbf{S}^{-1}\left(\overline{\mathbf{X}}-\boldsymbol{\mu}\right) \le F_{p,n-p}(\alpha)\right) = 1-\alpha$$

Sample Measurements: $\mathbf{x}_1, ..., \mathbf{x}_n$

$$\Rightarrow \quad \bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i \qquad \mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})'$$

$\Rightarrow \quad (1-\alpha)\,100\%$ Confidence Interval for $\boldsymbol{\mu}$:

All $\boldsymbol{\mu}$ such that $\left[\frac{n-p}{p\,(n-1)}\right] n\,(\bar{\mathbf{x}}-\boldsymbol{\mu})'\mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) \le F_{p,n-p}(\alpha)$

Equivalently: All $\boldsymbol{\mu}$ such that $(\bar{\mathbf{x}}-\boldsymbol{\mu})'\mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) \le \left[\frac{p(n-1)}{n(n-p)}\right] F_{p,n-p}(\alpha)$

Confidence Ellipsoid centered at $\bar{\mathbf{x}}$ with axes: $\pm\left[\sqrt{\lambda_i}\sqrt{\left[\frac{p(n-1)}{n(n-p)}\right] F_{p,n-p}(\alpha)}\right]\mathbf{e}_i$

where $\{\lambda_i\}$ and $\{\mathbf{e}_i\}$ are the eigenvalues and eigenvectors of $\mathbf{S}$.

Confidence Regions for Mean Vector - II

$$\left[\frac{n-p}{p(n-1)}\right] n\left(\overline{\mathbf{X}}-\boldsymbol{\mu}\right)'\mathbf{S}^{-1}\left(\overline{\mathbf{X}}-\boldsymbol{\mu}\right) \sim F_{p,n-p}$$

$$\Rightarrow \quad P\left(\left[\frac{n-p}{p(n-1)}\right] n\left(\overline{\mathbf{X}}-\boldsymbol{\mu}\right)'\mathbf{S}^{-1}\left(\overline{\mathbf{X}}-\boldsymbol{\mu}\right) \leq F_{p,n-p}(\alpha)\right) = 1-\alpha$$

In general $\forall$ fixed a:

$$P\left[\mathbf{a}'\overline{\mathbf{X}} - \sqrt{\left[\frac{p(n-1)}{n(n-p)}\right] F_{p,n-p}(\alpha)\ \mathbf{a}'\mathbf{S}\mathbf{a}} \leq \mathbf{a}'\mu \leq \mathbf{a}'\overline{\mathbf{X}} + \sqrt{\left[\frac{p(n-1)}{n(n-p)}\right] F_{p,n-p}(\alpha)\ \mathbf{a}'\mathbf{S}\mathbf{a}}\right] = 1-\alpha$$

Confidence Regions for Mean Vector - III

Sample Measurements: $\mathbf{x}_1, ..., \mathbf{x}_n$

$$\Rightarrow \quad \overline{\mathbf{x}} = \frac{1}{n}\sum_{j=1}^{n}\mathbf{x}_j \qquad \mathbf{S} = \frac{1}{n-1}\sum_{j=1}^{n}\left(\mathbf{x}_j - \overline{\mathbf{x}}\right)\left(\mathbf{x}_j - \overline{\mathbf{x}}\right)'$$

Defining: $\mathbf{a}_1{}' = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \quad ... \quad \mathbf{a}_p{}' = \begin{bmatrix} 0 & 0 & \cdots & 1 \end{bmatrix}$

$$\Rightarrow \quad \overline{x}_i \pm \sqrt{\left[\frac{p\,(n-1)}{(n-p)}\right] F_{p,n-p}\,(\alpha)}\;\sqrt{\frac{s_{ii}}{n}} \text{ will contain } \mu_i \text{ with confidence coefficient } 1-\alpha$$

For differences among means: $\mu_i - \mu_k$:
e.g. $\mu_1 - \mu_2$ with $\mathbf{a}' = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 \end{bmatrix}$:

$$(\overline{x}_i - \overline{x}_k) \pm \sqrt{\left[\frac{p\,(n-1)}{(n-p)}\right] F_{p,n-p}\,(\alpha)}\;\sqrt{\frac{s_{ii} - 2s_{ik} + s_{kk}}{n}} \text{ will contain } \mu_i - \mu_k \text{ w. confi coeff. } 1-\alpha.$$

$(1-\alpha)\,100\%$ Confidence Ellipsoid for (μ_i, μ_k) : Set of (μ_i, μ_k) such that:

$$n \begin{bmatrix} (\overline{x}_i - \mu_i) & (\overline{x}_k - \mu_k) \end{bmatrix} \begin{bmatrix} s_{ii} & s_{ik} \\ s_{ik} & s_{kk} \end{bmatrix}^{-1} \begin{bmatrix} (\overline{x}_i - \mu_i) \\ (\overline{x}_k - \mu_k) \end{bmatrix} \leq \sqrt{\left[\frac{p\,(n-1)}{(n-p)}\right] F_{p,n-p}\,(\alpha)}$$

Bonferroni Inequality: m Simultaneous Inferences (Tests/CI's):
$C_1, ..., C_m$ with $P(C_i \text{ False}) = \alpha_i$

$$P(\text{All } C_i \text{ True}) = 1 - P(\text{At least one } C_i \text{ False}) \geq 1 - \sum_{i=1}^{m} P(C_i \text{ False}) =$$

$$= 1 - \sum_{i=1}^{m} (1 - P(C_i \text{True})) = 1 - (\alpha_1 + ... + \alpha_m).$$

Setting $\alpha_i = \frac{\alpha}{m} \Rightarrow \quad P(\text{All } C_i \text{ True}) \geq 1 - m\left(\frac{\alpha}{m}\right) = 1 - \alpha$
$(1-\alpha)\,100\%$ Bonferroni Simultaneous Confidence Intervals for $\{\mu_i\}$:

$$\overline{x}_i \pm t_{n-1}\left(\frac{\alpha}{2p}\right)\sqrt{\frac{s_{ii}}{n}} \quad i = 1, ..., p$$

Large-Sample Inferences: $\mathbf{X}_1, ..., \mathbf{X}_n$ with $E\{\mathbf{X}_i\} = \boldsymbol{\mu}_X \quad V\{\mathbf{X}_i\} = \boldsymbol{\Sigma}_X$ positive def.
Sample measurements: $\mathbf{x}_1, ..., \mathbf{x}_n$ with large $n - p$:
Reject $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ if $n(\overline{\mathbf{x}} - \boldsymbol{\mu}_0)'\mathbf{S}^{-1}(\overline{\mathbf{x}} - \boldsymbol{\mu}_0) \geq \chi_p^2(\alpha)$

$$P\left(\mathbf{a}'\overline{\mathbf{X}} - \sqrt{\chi_p^2(\alpha)}\sqrt{\frac{\mathbf{a}'\mathbf{S}\mathbf{a}}{n}} \leq \mathbf{a}'\mu \leq \mathbf{a}'\overline{\mathbf{X}} + \sqrt{\chi_p^2(\alpha)}\sqrt{\frac{\mathbf{a}'\mathbf{S}\mathbf{a}}{n}}\right) \approx 1 - \alpha \quad \forall\, \mathbf{a}$$

$\Rightarrow$ Approximate $(1-\alpha)\,100\%$ CI for $\mathbf{a}'\mu$: $\quad \mathbf{a}'\overline{\mathbf{x}} \pm \sqrt{\chi_p^2(\alpha)}\sqrt{\frac{\mathbf{a}'\mathbf{S}\mathbf{a}}{n}}$.